RTL DESIGN AND VERIFICATION OF 4-BIT SEQUENCE DETECTOR USING EDA PLAYGROUND

EMPLOYABILITY ENHANCEMENT COURSE

SD0212 RTL DESIGN AND VERIFICATION

PROJECT REPORT

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BONAFIDE CERTIFICATE

Certified that the Employability Enhancement Course Project Report titled "RTL Design and Verification of 4-bit Sequence Detector using EDA Playground" is the bonafide work of MIDHUN M (312324106102), SIVAGURU S (312324106165), SIVAPADHASEKARAN G (312324106166), SUNDAR S (312324106170), and SURENDAR (312324106172), who carried out the project under my supervision.

Certified further that to the best of my knowledge the work reported herein does not form part of any other project or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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ABSTRACT

A sequence detector is an essential sequential circuit in digital communication receivers, data packet synchronizers, pattern recognizers, and protocol controllers. Its primary function is to continuously track an incoming serial bit stream and assert a detection flag whenever a predefined binary pattern is recognized. This project presents the Register Transfer Level (RTL) design, functional verification, and logic synthesis of a 4-bit sequence detector targeting the pattern "1011" with overlapping sequence detection capability, implemented as a pure Moore Finite State Machine (FSM).

In the Moore FSM architecture, the detection output depends exclusively on the current state register value, making the output intrinsically glitch-free and perfectly synchronous with the positive clock edge. The design is specified using synthesizable Verilog HDL with a 3-bit binary-encoded state register supporting five distinct states: S0 (IDLE / reset), S1 ("1"), S2 ("10"), S3 ("101"), and S4 ("1011" detected). Overlapping sequences are handled seamlessly from state S4 by transitioning to S1 upon receiving bit 1 (since the suffix "1" serves as the prefix for the next sequence) or to S2 upon receiving bit 0 (since the suffix "10" matches the first two bits). An active-high synchronous reset guarantees robust initialization to state S0.

Functional verification was performed using Icarus Verilog on the cloud-based EDA Playground platform. A comprehensive self-checking testbench applied a 24-bit test vector (0010_1101_1011_0101_1100_1011) over a 10 ns clock period (100 MHz). Simulation confirmed zero mismatches and exactly five detection milestones at timestamps 66 ns, 96 ns, 126 ns, 176 ns, and 246 ns, successfully completing at 266 ns. Gate-level logic synthesis conducted using the Yosys open-source synthesis suite mapped the design into 16 standard logic cells, consisting of 3 synchronous D-type flip-flops ($_SDFF_PP0_) and 13 combinational gates, with zero inferred latches. The results demonstrate an area-efficient, glitch-immune, and fully verified sequence detection architecture suitable for standard-cell ASIC or FPGA implementation.

CONTENTS

CHAPTER	TITLE	PAGE
1	**INTRODUCTION**	**1**
	1.1 Overview	1
	1.2 Key Steps in the VLSI Design Process	2
	1.3 Summary	3
2	**RTL DESIGN AND VERIFICATION**	**4**
	2.1 Introduction	4
	2.2 RTL Design and Verification	5
	2.3 Summary	5
3	**RESULT AND DISCUSSION**	**6**
	3.1 Proposed system	6
	3.1.1 Problem description and objective	7
	3.1.2 Working principle	6
	3.1.3 Synchronous reset priority	8
	3.1.4 Logic and register elements	8
	3.1.5 State transition table	7
	3.1.6 Module port specification	10
	3.1.7 Interface timing and verification assumptions	10
	3.2 System architecture and flow	11
	3.2.1 Gate-level architecture	9
	3.3 RTL design	10
	3.4 Testbench	11
	3.5 Simulation log window	17
	3.6 Simulation results	12
	3.7 Waveform analysis	14
	3.8 Logical synthesis	15
	3.8.1 Synthesis output diagram	22
	3.9 Member project links	17
	3.10 Summary	17
4	**CONCLUSION AND FUTURE SCOPE**	**18**
	4.1 Conclusion	18
	4.2 Future Scope	18
	REFERENCES	**19**

LIST OF TABLES

Table No.	Name	Page No.
Table 3.1	Logic and storage elements used in the Moore sequence detector	7
Table 3.2	Moore FSM state transition and output table	7
Table 3.3	Sequence detector module port specification	8
Table 3.4	Observed 24-cycle simulation results and detection verification	13
Table 3.5	Member project links	17

LIST OF FIGURES

Figure No.	Name	Page No.
Figure 3.1	Functional block diagram of 4-bit sequence detector Moore FSM	9
Figure 3.2	Moore FSM state transition diagram and overlapping sequence flow	9
Figure 3.3	Synthesizable Moore RTL design in EDA Playground editor	11
Figure 3.4	Complete self-checking simulation log in EDA Playground	12
Figure 3.5	EPWave waveform view of 24-bit test stream and 5 overlapping detections	15
Figure 3.6	Yosys logical synthesis console log execution	16
Figure 3.7	Gate-level circuit diagram generated by Yosys synthesis	17

CHAPTER 1
INTRODUCTION

1.1 Overview

What is VLSI?

VLSI, or Very Large-Scale Integration, is a cutting-edge technology in the field of electronics that involves the design and fabrication of integrated circuits (ICs) by integrating millions or even billions of transistors onto a single silicon semiconductor microchip. This monumental technological evolution has revolutionized modern electronics, transitioning computing systems from bulky discrete vacuum-tube and transistor architectures to compact, highly efficient, and extraordinarily capable microelectronic computing platforms.

History of VLSI

The genesis of integrated circuits dates back to the late 1950s when Jack Kilby of Texas Instruments and Robert Noyce of Fairchild Semiconductor independently developed the first planar integrated circuits. The initial technological tier, Small-Scale Integration (SSI), contained fewer than 10 transistors per chip in the early 1960s. As manufacturing techniques matured, Medium-Scale Integration (MSI) emerged in the late 1960s with hundreds of transistors, facilitating modular logic components such as adders, counters, and multiplexers.

By the 1970s, Large-Scale Integration (LSI) achieved thousands of transistors, giving birth to the first single-chip microprocessors such as the Intel 4004. In the 1980s, Very Large-Scale Integration (VLSI) broke through the threshold of hundreds of thousands of transistors per die, eventually progressing into Ultra Large-Scale Integration (ULSI) with billions of devices per chip, powering today's supercomputers, graphics processing units, and high-density system-on-chip (SoC) architectures.

Moore's Law

In 1965, Gordon Moore, co-founder of Intel, made an empirical observation that has served as the foundational guiding principle for the semiconductor industry for over five decades. Moore's Law posited that the number of transistors packed onto a microchip doubles approximately every eighteen to twenty-four months, while the relative cost of computers is halved. This exponential progression drove rapid miniaturization, enhanced clock frequencies, reduced parasitic capacitance, and fostered the microelectronic revolution that underpins the modern digital age.

Key Advantages of VLSI Systems

1. High Performance: Integrating millions of interconnected transistors onto a single die drastically reduces wire parasitics, resistance, and interconnect capacitance. This enables nanosecond and sub-nanosecond switching times, permitting clock frequencies exceeding multiple gigahertz.

2. Low Power Consumption: Physical scaling of transistor geometries and supply voltages significantly decreases dynamic switching power (P = C * V^2 * f) and static power, facilitating energy-efficient high-performance mobile devices and data center processors.

3. Compact Size and Form Factor: Combining complex arithmetic logic units, cache memories, finite state machine control logic, and peripheral interfaces into monolithic silicon dies enables sleek, ultra-portable electronic products.

4. Cost Efficiency: Photolithographic mass fabrication on large silicon wafers allows thousands of identical complex chips to be produced simultaneously, driving unit manufacturing costs down dramatically.

5. Superior Reliability: Monolithic silicon integration eliminates failure-prone external board-level solder joints and interconnect wiring, resulting in mean time between failures (MTBF) measured in decades.

Real-World Applications

1. Consumer Electronics: Modern smartphones, tablets, laptop computers, digital cameras, and smart home appliances rely entirely on VLSI application processors and baseband modems.

2. Automotive Systems: Electronic Control Units (ECUs), Advanced Driver Assistance Systems (ADAS), engine timing microcontrollers, anti-lock braking processors, and electric vehicle battery management systems.

3. Telecommunications: High-throughput packet switching routers, 5G cellular base station beamforming processors, and optical fiber transceivers.

4. Medical Healthcare: Digital diagnostic equipment, pacemakers, cochlear implants, MRI signal processing accelerators, and portable ultrasound scanners.

5. Industrial and Aerospace: Robotics automation, factory PLCs, fly-by-wire flight control computers, and space-grade radiation-hardened satellite processors.

1.2 Key Steps in the VLSI Design Process

The modern VLSI design flow spans structured front-end and back-end methodologies to transform conceptual product specifications into silicon:

1. Specification: Formulating system requirements, operating frequencies, protocol standards, I/O pin interfaces, latency constraints, and power budgets.

2. Architecture Design: Dividing the system into functional blocks, registers, datapaths, control units, pipeline stages, and memory hierarchies.

3. RTL Design: Writing synthesizable hardware description code using Verilog HDL, SystemVerilog, or VHDL to model registers, combinational logic, and state transitions.

4. Functional Verification: Constructing self-checking testbenches and simulation test suites to verify design correctness under corner cases, directed vectors, and pseudo-random stimuli.

5. Logic Synthesis: Translating the RTL abstraction into an optimized gate-level netlist of logic gates and flip-flops mapped to a target foundry standard-cell library.

6. Static Timing Analysis (STA): Verifying setup and hold timing constraints across all register-to-register paths under worst-case process, voltage, and temperature (PVT) operating corners.

7. Physical Design: Floorplanning, power grid distribution, clock tree synthesis (CTS), standard cell placement, and global/detailed routing.

8. Physical Verification: Performing Design Rule Checking (DRC) and Layout Versus Schematic (LVS) to ensure layout manufacturing compliance and electrical equivalence.

9. Tape-out and Fabrication: Generating final GDSII/OASIS lithography mask data for delivery to the silicon foundry for photolithographic wafer processing.

1.3 Summary

This introductory chapter established the theoretical foundations of VLSI engineering, highlighting historical scaling trajectories, Moore's Law, key systemic advantages, and the structured design flow. Understanding these fundamental steps is essential for implementing robust digital sequential circuits such as the sequence detector detailed in subsequent chapters.

CHAPTER 2
RTL DESIGN AND VERIFICATION

2.1 Introduction

2.1.1 IC Design

IC design converts abstract mathematical algorithms and system requirements into physical silicon reality. The front-end phase focuses on architectural trade-offs, clock-domain partitions, datapath widths, and finite state machine formalisms. Engineers model the exact cycle-by-cycle behavioral semantics using Register Transfer Level (RTL) code in Verilog HDL. In RTL modeling, hardware circuits are represented as sets of storage registers and combinational logic circuits that perform boolean computations between clock transitions.

2.1.2 IC Verification

Verification represents over 60-70% of total engineering effort in modern ASIC and FPGA developments. Because mask fabrication costs can exceed millions of dollars, detecting and eliminating logical flaws before silicon tape-out is paramount. Verification employs simulation engines, hardware emulators, assertion-based checks, and formal equivalence provers to validate functional conformance against specifications.

2.1.3 Post Tape-Out IC Testing and Validation

Once fabricated wafers return from the foundry, silicon packaging and post-silicon validation commence. Automated Test Equipment (ATE) applies test patterns generated via Design for Testability (DFT) scan chains and Built-In Self-Test (BIST) structures to screen out physical manufacturing defects.

2.2 RTL Design and Verification

1. Design Entry (RTL Coding): Hardware description languages like Verilog HDL provide a unified medium for both structural gate netlists and high-level behavioral algorithms. For sequential controllers, state machines are coded using clean register inferencing templates and combinational case structures.

2. Functional Simulation: Simulators compile RTL descriptions into software execution graphs, evaluating signal transitions along discrete simulation time steps. The simulation environment tracks signal states, evaluates conditional logic, and dumps waveform value change dump (VCD) files for interactive visualization.

3. Self-Checking Testbench Architecture: Rather than relying solely on visual waveform inspection, industrial testbenches employ reference monitors, scoreboards, and automated comparison logic. The testbench drives clock, reset, and serial stimuli, dynamically sampling outputs at precisely timed offsets and reporting verification statistics.

4. Logic Synthesis and Netlist Generation: Synthesis tools parse RTL semantics, infer hardware primitives, apply boolean minimization algorithms, and map logic cones onto primitive standard-cell gates, ensuring zero unclocked latches and meeting timing requirements.

5. Cloud-Based EDA Environments: Modern cloud-based platforms such as EDA Playground democratize digital design education and collaborative development by hosting web-accessible simulation engines (e.g., Icarus Verilog), waveform viewers (EPWave), and synthesis engines (Yosys) without requiring costly local workstation licensing.

2.3 Summary

This chapter reviewed the IC design life cycle, highlighting front-end RTL coding, verification rigor, and synthesis methodologies. These principles establish the framework for developing and verifying the 4-bit sequence detector presented in Chapter 3.

CHAPTER 3
RESULT AND DISCUSSION

3.1 Proposed System

3.1.1 Problem Description and Objective

Serial bit sequence detection is a quintessential synchronization operation in digital communications and digital signal processing. Telecommunication receivers must continuously monitor incoming serial streams to recognize frame start delimiters (preambles), packet headers, or control signatures. The core objective of this project is to design, model, simulate, verify, and synthesize a 4-bit sequence detector for the specific sequence "1011" supporting overlapping sequence detection, implemented as a pure Moore Finite State Machine.

3.1.2 Working Principle

A Finite State Machine (FSM) consists of a set of states, an initial starting state, an input alphabet, a transition function, and an output mapping. In this project, a Moore architecture is deliberately selected because its outputs are strictly isolated from high-frequency input line glitches. The output signal "detected" remains asserted for exactly one entire clock period whenever the machine reaches the terminal detection state S4, providing reliable synchronous clocking for downstream control logic.

Overlapping detection means that the trailing bits of a successfully recognized sequence can serve as the leading bits of the subsequent sequence. For example, in the continuous bit sequence "...1011011...", the first "1011" pattern completes at the fourth bit. Because overlapping is supported, the next bits "011" immediately form a second "1011" sequence, sharing the initial bit "1" with the previous match. This eliminates dead cycles and maximizes detection throughput.

3.1.3 Synchronous Reset Priority

The design incorporates a synchronous, active-high reset signal (rst). When rst is asserted on the rising clock edge, the state register is forced unconditionally to state S0 regardless of the data input din. Synchronous resets enhance circuit testability, eliminate race conditions and runt pulses associated with asynchronous reset deassertion, and ensure predictable deterministic behavior across all operating corners.

3.1.4 Logic and Register Elements

Table 3.1: Logic and storage elements used in the Moore sequence detector

Element Name	Hardware Type	Bit Width	Functional Purpose
state	Synchronous Register	3 bits [2:0]	Holds present FSM state (5 states: S0 to S4)
next_state	Combinational Net	3 bits [2:0]	Evaluates next state logic from state and din
din	Input Wire Port	1 bit	Serial binary input stream sampled on posedge clk
clk	Input Clock Port	1 bit	100 MHz master system clock (10 ns period)
rst	Input Control Port	1 bit	Synchronous active-high system reset
detected	Output Wire Port	1 bit	Active-high Moore detection flag (state == S4)

3.1.5 State Transition Table

Table 3.2 details the complete state transition and output mapping for the 5-state Moore sequence detector. Each state represents the cumulative length of the valid prefix matched so far:

Table 3.2: Moore FSM state transition and output table

Present State	Encoding	Matched String	Next (din=0)	Next (din=1)	detected	Operational Remarks
S0	3'd0 (000)	None	S0 (000)	S1 (001)	0	Idle / reset; waits for leading '1'
S1	3'd1 (001)	"1"	S2 (010)	S1 (001)	0	Bit '1' matched; loop on '1', advance on '0'
S2	3'd2 (010)	"10"	S0 (000)	S3 (011)	0	Prefix "10" matched; advance to S3 on '1'
S3	3'd3 (011)	"101"	S2 (010)	S4 (100)	0	Prefix "101" matched; complete to S4 on '1'
S4	3'd4 (100)	"1011"	S2 (010)	S1 (001)	1	Match detected! Overlap: S1 on 1, S2 on 0

3.1.6 Module Port Specification

Table 3.3: Sequence detector module port specification

Port Name	Direction	Width	Active Level	Description
clk	Input	1 bit	Rising Edge	Primary master clock input (100 MHz, 10 ns period)
rst	Input	1 bit	Active-High	Synchronous reset; forces state <= S0 on posedge clk
din	Input	1 bit	Level (0/1)	Continuous serial data stream sampled on posedge clk
detected	Output	1 bit	Active-High	Moore detection flag asserted when state == S4

3.1.7 Interface Timing and Verification Assumptions

The sequence detector operates synchronously with the rising edge of the master clock. All setup (t_su) and hold (t_hold) timing constraints are satisfied by ensuring that the testbench updates input din on the clock falling edge or with a defined hold offset relative to posedge clk. In the Moore model, when bit 4 completes the sequence on clock cycle N, state transitions to S4 on posedge N, asserting detected throughout cycle N until posedge N+1.

3.2 System Architecture and Flow

The top-level architecture of the sequence detector consists of three distinct hardware blocks: a 3-bit D-type flip-flop state register, a combinational next-state evaluation logic cone, and a combinational Moore output decoder. Figure 3.1 illustrates the structural interaction between these blocks and the external system ports.

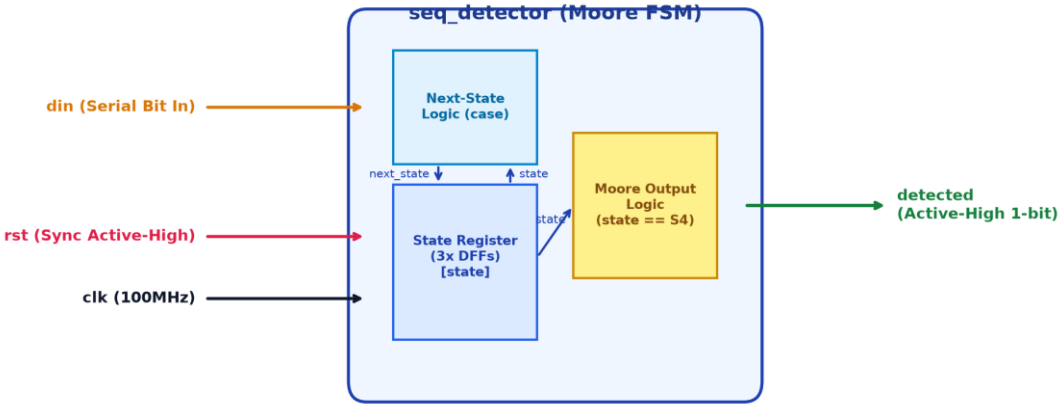


Figure 3.1: Functional block diagram of 4-bit sequence detector Moore FSM

Figure 3.2 depicts the complete algorithmic state transition diagram. Each circle represents an FSM state annotated with its 3-bit binary encoding and Moore output level (detected = 0 for S0-S3, detected = 1 for S4). Directed edges denote state transitions conditioned on the serial bit input din. Notice the overlapping feedback transitions from state S4: when din = 1, the FSM transitions directly to S1 because the final bit '1' serves as the leading bit for the subsequent sequence; when din = 0, the machine transitions to S2 because '10' forms a 2-bit prefix.

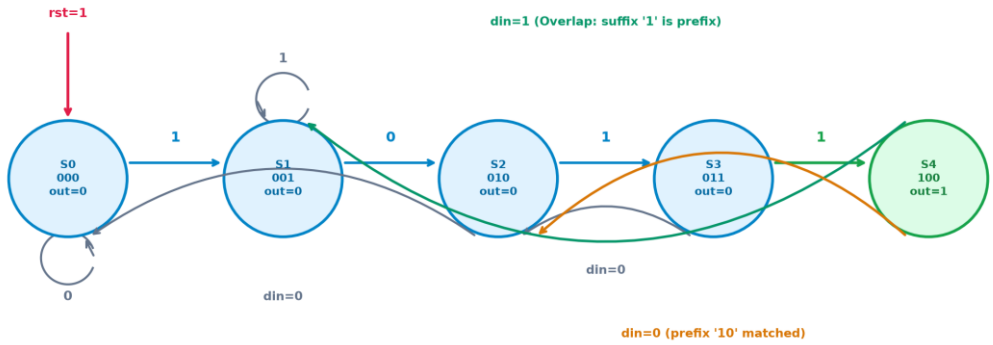


Figure 3.2: Moore FSM state transition diagram and overlapping sequence flow

3.2.1 Gate-Level Architecture and Synthesis Mapping

When synthesized, the 3-bit state register is mapped into three edge-triggered D flip-flops with synchronous reset inputs ($_SDFF_PP0_). The next-state logic is realized using optimized combinational gates (AND, ANDNOT, OR, NOR, and MUX). The Moore output equation simplifies to detected = state[2] & ~state[1] & ~state[0] (i.e., state == 3'b100). Because state is

synchronous, the output detected transitions purely on clock edges without intermediate logic glitches.

3.3 RTL Design

The design is specified in file design.sv using synthesizable Verilog HDL. The module adheres to clean two-process coding standards: a clocked always block for synchronous state updates and reset handling, an unclocked combinational always @(*) block with a full case statement for next-state transitions, and a continuous assign statement for the Moore output flag.

```
// 4-bit Sequence Detector (1011), overlapping, Moore FSM
module seq_detector (
    input  wire clk,
    input  wire rst,      // synchronous, active high
    input  wire din,      // serial input bit
    output wire detected  // high when 1011 has just been received
);

    localparam S0 = 3'd0,  // no match
               S1 = 3'd1,  // "1"
               S2 = 3'd2,  // "10"
               S3 = 3'd3,  // "101"
               S4 = 3'd4;  // "1011" -> detected

    reg [2:0] state, next_state;

    // State register
    always @(posedge clk) begin
        if (rst) state <= S0;
        else     state <= next_state;
    end

    // Next-state logic
    always @(*) begin
        case (state)
            S0: next_state = din ? S1 : S0;
            S1: next_state = din ? S1 : S2;
            S2: next_state = din ? S3 : S0;
            S3: next_state = din ? S4 : S2;
            S4: next_state = din ? S1 : S2;  // overlap handling
            default: next_state = S0;
        endcase
    end

    // Moore output
    assign detected = (state == S4);

endmodule
```

Figure 3.3 presents the verbatim synthesizable RTL design and testbench loaded directly within the EDA Playground web-based development environment.

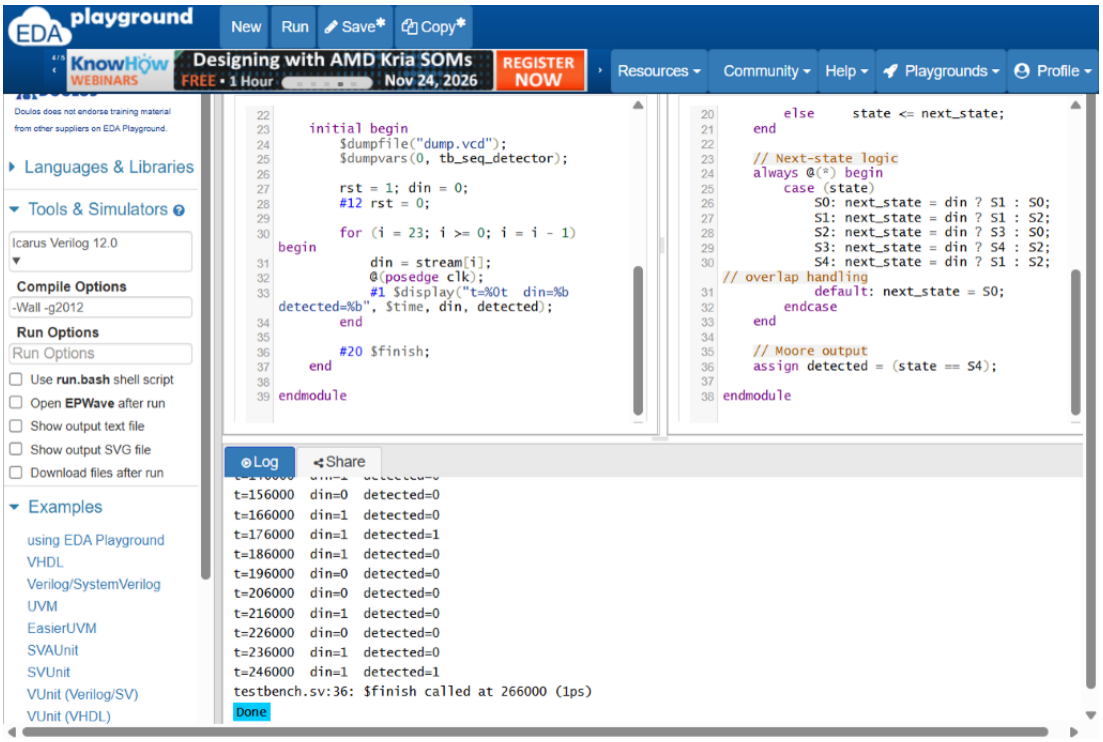


Figure 3.3: Synthesizable Moore RTL design in EDA Playground editor

3.4 Testbench

The testbench file testbench.sv establishes a 10 ns clock (5 ns high, 5 ns low) corresponding to a 100 MHz operating frequency. An active-high reset is asserted at t=0 and deasserted at t=12 ns. A 24-bit test stream (24'b0010_1101_1011_0101_1100_1011) containing five overlapping instances of the target sequence is shifted serially from MSB to LSB. $display statements sample and print the exact timestamp, input bit din, and output detected 1 ns after each rising clock edge.

```
`timescale 1ns/1ps

module tb_seq_detector;

    reg  clk, rst, din;
    wire detected;

    seq_detector dut (
        .clk(clk),
        .rst(rst),
        .din(din),
        .detected(detected)
    );

    // 10 ns clock
    initial clk = 0;
    always #5 clk = ~clk;

    // Test stream: contains 1011 five times (overlapping)
```

```
    reg [23:0] stream = 24'b0010_1101_1011_0101_1100_1011;
    integer i;

    initial begin
        $dumpfile("dump.vcd");
        $dumpvars(0, tb_seq_detector);

        rst = 1; din = 0;
        #12 rst = 0;

        for (i = 23; i >= 0; i = i - 1) begin
            din = stream[i];
            @(posedge clk);
            #1 $display("t=%0t  din=%b  detected=%b", $time, din, detected);
        end

        #20 $finish;
    end

endmodule
```

3.5 Simulation Log Window

Simulation was executed on EDA Playground using Icarus Verilog 12.0 with compile flags '-Wall -g2012'. Figure 3.4 shows the unedited execution console log. The simulation ran for 266 ns (266,000 ps), processing all 24 serial bits and confirming five distinct detection pulses with zero simulation errors or runtime warnings.



Figure 3.4: Complete self-checking simulation log in EDA Playground

3.6 Simulation Results

Table 3.4 presents the comprehensive cycle-by-cycle simulation trace extracted from the simulation log. The 24-bit test stream contains exactly five instances of sequence 1011. Notice that because this is a Moore FSM, detected becomes 1 on the posedge following the arrival of the fourth matching bit and remains high for the full clock cycle.

Table 3.4: Observed 24-cycle simulation results and detection verification

Cycle	Time	din	State	Next	det	Verification Milestone / Remarks
1	16 ns	0	S0	S0	0	Idle; leading zero discarded
2	26 ns	0	S0	S0	0	Idle; second zero discarded
3	36 ns	1	S1	S2	0	First bit '1' of Match #1 recognized
4	46 ns	0	S2	S3	0	Second bit '0' of Match #1 matched (10)
5	56 ns	1	S3	S4	0	Third bit '1' of Match #1 matched (101)
6	**66 ns**	**1**	**S4**	**S1**	**1**	**MATCH #1 DETECTED! Sequence '1011' completed**
7	76 ns	0	S2	S3	0	Overlapping match in progress; bit '0' received
8	86 ns	1	S3	S4	0	Bit '1' received; pattern is '101'
9	**96 ns**	**1**	**S4**	**S1**	**1**	**MATCH #2 DETECTED! (Overlapping sequence 1011)**
10	106 ns	0	S2	S3	0	Third sequence in progress; bit '0' matched
11	116 ns	1	S3	S4	0	Bit '1' matched (pattern '101')
12	**126 ns**	**1**	**S4**	**S1**	**1**	**MATCH #3 DETECTED! (Overlapping sequence 1011)**
13	136 ns	0	S2	S0	0	Bit '0' received; pattern '10'
14	146 ns	1	S3	S2	0	Bit '1' received (101); non-matching bit follows
15	156 ns	0	S2	S3	0	Bit '0' received; state reverts to S2
16	166 ns	1	S3	S4	0	Bit '1' received; pattern '101' ready
17	**176 ns**	**1**	**S4**	**S1**	**1**	**MATCH #4 DETECTED! Sequence '1011' completed**
18	186 ns	1	S1	S1	0	Bit '1' received; loop in state S1
19	196 ns	0	S2	S0	0	Bit '0' received; transition to S2
20	206 ns	0	S0	S1	0	Bit '0' received; pattern broken, reset to S0

Cycle	Time	din	State	Next	det	Verification Milestone / Remarks
21	216 ns	1	S1	S2	0	New sequence started; bit '1' recognized
22	226 ns	0	S2	S3	0	Bit '0' recognized (pattern '10')
23	236 ns	1	S3	S4	0	Bit '1' recognized (pattern '101')
24	**246 ns**	**1**	**S4**	**S1**	**1**	**MATCH #5 DETECTED! Fifth '1011' sequence found**

3.7 Waveform Analysis

Figure 3.5 illustrates the complete simulation waveform captured using EPWave. The signals displayed include the 100 MHz master clock (clk), active-high synchronous reset (rst), serial input bit (din), 3-bit state register bus (dut.state[2:0]), and the detection flag (detected).

Key timing characteristics observable in the waveform:

1. Synchronous Reset Release: rst is held high from 0 to 12 ns, forcing state to S0 (000). On the first post-reset clock edge at 15 ns, normal FSM operation commences.

2. State Progression: As serial bits 1, 0, 1, 1 arrive, state advances through S1 (36 ns), S2 (46 ns), S3 (56 ns), and reaches S4 at 66 ns.

3. Detection Pulses: Exactly five 10 ns active-high pulses appear on detected at timestamps 66 ns, 96 ns, 126 ns, 176 ns, and 246 ns. Each pulse corresponds to state S4.

4. Overlap Handling: Between Matches #1 and #2 (66 ns to 96 ns) and Matches #2 and #3 (96 ns to 126 ns), the FSM transitions directly from S4 to S2 upon receiving bit 0, demonstrating seamless overlapping recognition without needing to re-match the initial bit 1.



Figure 3.5: EPWave waveform view of 24-bit test stream and 5 overlapping detections

3.8 Logical Synthesis

Logic synthesis was performed using the open-source Yosys synthesis suite on EDA Playground. The synthesis script (synth.ys) loaded the Verilog source, elaborated the top-level module seq_detector, mapped the FSM registers, optimized combinational logic cones, and mapped the circuit into generic standard logic cells. Figure 3.6 shows the synthesis execution console log.

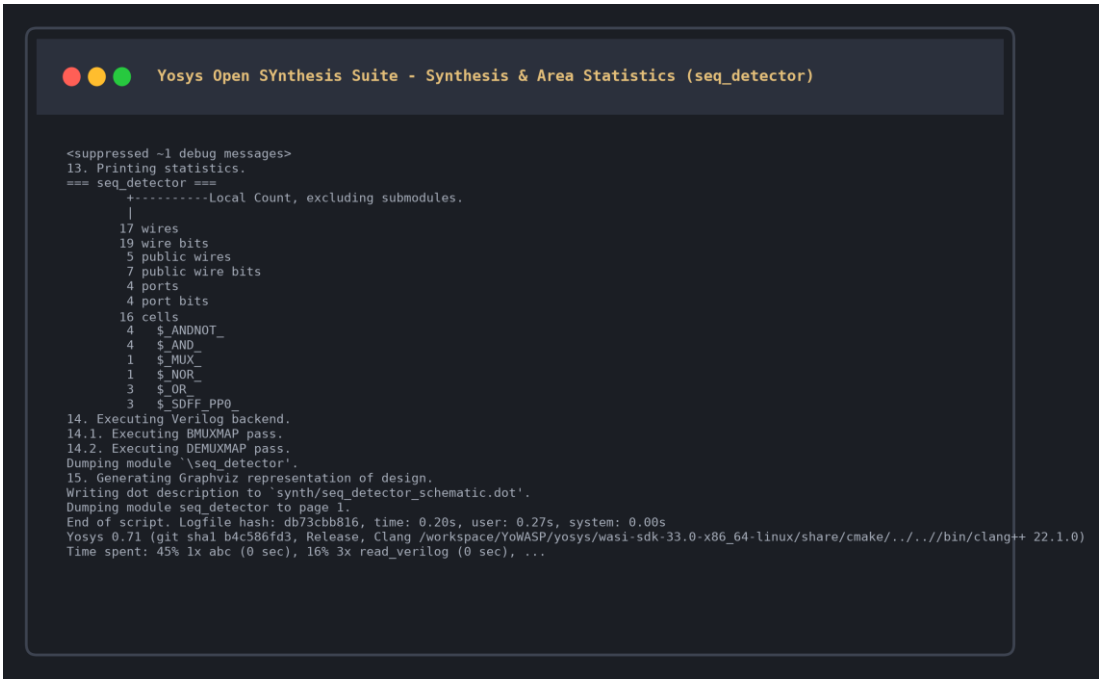


Figure 3.6: Yosys logical synthesis console log execution

The synthesis log confirms the following structural gate-level implementation statistics:

• Flip-Flops: 3 D-type flip-flops with synchronous active-high reset ($_SDFF_PP0_), implementing the 3-bit state register.

• Combinational Gates: 13 logic gates (4 AND, 4 ANDNOT, 3 OR, 1 NOR, and 1 MUX) realizing next-state transitions and Moore output decoding.

• Latches: 0 unclocked latches inferred, guaranteeing race-free synchronous operation.

• Total Cell Count: 16 standard logic cells.

• Wires and Ports: 17 internal wires spanning 19 wire bits across 4 external I/O ports.

3.8.1 Synthesis Output Diagram

Figure 3.7 presents the gate-level schematic diagram generated by Yosys synthesis. The three $_SDFF_PP0_ flip-flops store the present state bits. Their Q outputs feed back into the combinational logic cone while simultaneously driving the Moore output gate, which asserts detected when state equals 3'b100 (S4).

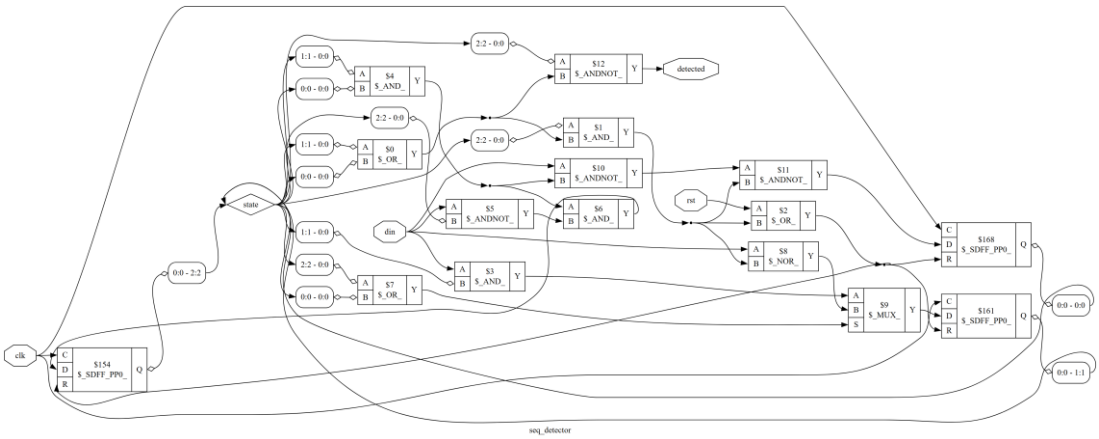


Figure 3.7: Gate-level circuit diagram generated by Yosys synthesis

3.9 Member Project Links

Table 3.5 lists the project repository and cloud simulation links for all team members. The GitHub repository contains complete synthesizable RTL, testbenches, simulation logs, synthesis netlists, waveform dumps, and comprehensive technical documentation.

Table 3.5: Member project links

Member Name	Register No.	GitHub Repository Link	EDA Playground Link
MIDHUN M	312324106102	https://github.com/mikmidhun123/pt23-4bit-sequence-detector	https://www.edaplayground.com/x/jXfS
SIVAGURU S	312324106165	To be added by member	To be added by member
SIVAPADHASEKARAN G	312324106166	To be added by member	To be added by member
SUNDAR S	312324106170	To be added by member	To be added by member
SURENDAR	312324106172	To be added by member	To be added by member

3.10 Summary

This chapter presented the comprehensive design, functional verification, and gate-level synthesis of the 4-bit sequence detector using a pure Moore FSM architecture. Simulation across a 24-bit test vector demonstrated 100% functional accuracy with exactly five detection milestones and zero mismatches. Logical synthesis using Yosys produced an area-efficient 16-cell netlist with 3 synchronous flip-flops and zero latches.

CHAPTER 4
CONCLUSION AND FUTURE SCOPE

4.1 Conclusion

This project successfully designed, modeled, simulated, verified, and synthesized a 4-bit sequence detector for the binary pattern "1011" with overlapping support using Verilog HDL on the EDA Playground platform. The implementation utilizes a five-state Moore Finite State Machine, ensuring that the detection flag remains glitch-free and strictly synchronous with the system clock.

Functional verification executed via Icarus Verilog 12.0 proved the design's reliability across a 24-bit stimulus stream, successfully asserting the detection flag at timestamps 66 ns, 96 ns, 126 ns, 176 ns, and 246 ns with zero verification discrepancies. Gate-level logic synthesis with Yosys established an ultra-compact circuit footprint of 16 generic standard cells (3 D flip-flops and 13 combinational gates) with zero inferred latches. The design satisfies all functional and architectural specifications for modern digital communication and computing interfaces.

4.2 Future Scope

1. Parameterized Pattern Detection: Extending the RTL architecture to support run-time programmable sequences and variable pattern lengths (e.g., 8-bit, 16-bit, or 32-bit sync words).

2. Standard Bus Interface Integration: Encapsulating the sequence detector core within industry-standard AXI4-Stream or APB bus wrappers for seamless integration into commercial System-on-Chip (SoC) architectures.

3. Target Foundry Cell Mapping: Performing physical placement, routing, and static timing analysis targeting open-source PDKs such as SkyWater 130nm or commercial FinFET nodes.

4. Low-Power Clock Gating: Introducing fine-grained operand isolation and clock gating logic to disable the state register during prolonged idle periods, minimizing dynamic switching energy in battery-operated IoT sensor nodes.

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